

# Dani Roytburg

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**Summary** Master’s student in the Machine Learning Department at Carnegie Mellon University focused on reliable and interpretable AI, multi-agent training paradigms, and activation-based model analysis.

**Education**

|      |                                                                                                            |      |
|------|------------------------------------------------------------------------------------------------------------|------|
| M.S. | <b>Carnegie Mellon University, School of Computer Science</b><br>Machine Learning                          | 2027 |
| B.S. | <b>Emory University, College of Arts and Science</b><br>Computer Science; Quantitative and Social Sciences | 2025 |

**Positions**

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| <b>ML Alignment and Theory Scholars</b><br>Berkeley, CA | Research Fellow (incoming)<br>May 2026-September 2026 |
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Research fellowship advised by Shi Feng.

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| <b>Carnegie Mellon University</b><br>Pittsburgh, PA | Research Assistant, DI Group<br>September 2025-present |
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Develop multi-agent reinforcement learning environments to improve legibility of reasoning traces. Design user studies and adversarial setups to measure influence of legibility on human and multi-agent interactions.

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|----------------------------------------------|-------------------------------------------|
| <b>Martian Research</b><br>San Francisco, CA | Research Fellow<br>May 2025-February 2026 |
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Engineered activation profiles of language model evaluators to suppress self-preferential bias, applied to LLM routers (ICML 2026, MechInterp @ NeurIPS 2025). Winner of Mechanistic Router Interpretability Hackathon (40 submissions).

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| <b>Emory University</b><br>Atlanta, GA | Research Assistant, Digital Humanities Lab<br>November 2021-May 2025 |
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Designed and trained large-scale graph-text models; evaluated adversarial robustness of LLMs; built historical correspondence networks and dashboards.

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| <b>ZS Associates</b><br>Evanston, IL | Internship - Business Technology Solutions<br>June 2024-August 2024 |
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Automated ETL pipelines and implemented synthetic-data imputation for compromised client data.

**Cloverpop**  
Chicago, IL

Intern — Machine Learning  
January 2024-May 2024

Deployed decision-management systems and structured-prediction models on meeting transcripts.

**McCormick and Company**  
Hunt Valley, MD

Intern — Global Marketing and Sales Data Science  
June 2023-August 2023

Built Amazon sales data visualizations and automated analytics with GCP BigQuery and BERT-based embedding search.

**Northeastern University**  
Boston, MA

Research Assistant, Mediacloud Lab  
June 2022-October 2022

Extended NER to long-tail ethnics cuisines using Bon Appetit data.

## Publications

### Conference Papers

Dani Roytburg, Matthew Bozoukov, Matthew Nguyen, Jou Barzdukas, Mackenzie Puig-Hall, and Narmeen Oozeer. Are llm evaluators really narcissists? sanity checking self-preference evaluations. In *Forty-Third International Conference on Machine Learning*, 2026. URL: <https://arxiv.org/abs/2601.22548>

Dani Roytburg, Shreya Sridhar, and Daphne Ippolito. Measuring weak-to-strong legibility of reasoning models. In *ICML 2026 Workshop on AI for Good (AI4GOOD)*, 2026. URL: <https://openreview.net/forum?id=MRStw9vqEg>

Dani Roytburg and Beck Miller. Mind the gap: Pathways towards unifying ai safety and ethics research. In *Proceedings of the International Association for Safe and Ethical AI*, 2026. URL: <https://arxiv.org/abs/2512.10058>

Dani Roytburg, Matthew Bozoukov, Hongyu Fu, Matthew Nguyen, Jou Barzdukas, and Narmeen Fatimah Oozeer. Breaking the mirror: Activation-based mitigation of self-preference in llm evaluators. In *Mechanistic Interpretability Workshop at NeurIPS 2025*, 2025. Also accepted at: NeurIPS 2025 Workshop on Evaluating the Evolving LLM Lifecycle; NeurIPS 2025 Workshop on Reliable ML from Unreliable Data. URL: <https://arxiv.org/abs/2509.03647>

Dani Roytburg, Deborah Olorunisola, Sandeep Soni, and Lauren Klein. Words and action: Modeling linguistic leadership in # blacklivesmatter communities. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 19, pages 1690-1703, 2025. URL: <https://ojs.aaai.org/index.php/ICWSM/article/view/35895>

### Thesis

Daniel Roytburg. Generative argument mining: Pretrained language models are argumentative text parsers. Undergraduate thesis, Emory University, 2025. URL: <https://etd.library.emory.edu/concern/etds/5t34sm11w?locale=en>

**Recognition****Awards**

Best Poster, LTI Student Research Symposium, 2026

Emory University Dean's List, 2022, 2024-2025

Winner, Martian Research Mechanistic Interpretability Hackathon, 2025

**Skills**

Python Java R JavaScript Typescript SQL PyTorch JAX HuggingFace Transformers scikit-learn  
spaCy networkx D3.js React.js GCP Docker MySQL